



Q: Over \$1.8B committed to MegaETH public sale?



Ground Truth: NO

Standard Multi-Agent Debate



Same Evidence ε (all docs)



Round 1 (independent)

$$\cdot p_1^{(1)} = 0.99 \quad \cdot p_2^{(1)} = 0.94 \quad \cdot p_3^{(1)} = 0.97$$

Round 2 (after deliberation)

$$\cdot p_1^{(2)} = 0.24 \quad \cdot p_2^{(2)} = 0.92 \quad \cdot p_3^{(2)} = 0.86$$



$\hat{p} = 0.67$ (Brier = 0.45) $\rightarrow$ YES



Issue: All agents saw the same evidence.
They misled each other.

InfoDelphi (Ours)



Public Evidence ε^{pub} (shared context)



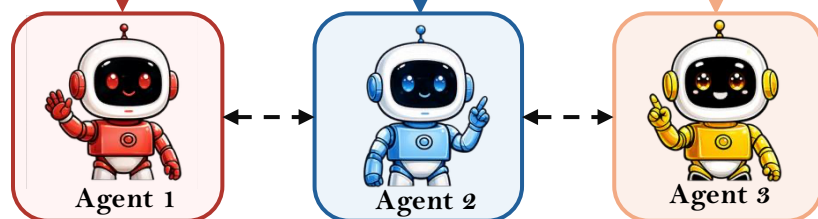
$\varepsilon_1^{private}$



$\varepsilon_2^{private}$



$\varepsilon_3^{private}$



Round 1 (independent)

$$\cdot p_1^{(1)} = 0.95 \quad \cdot p_2^{(1)} = 0.01 \quad \cdot p_3^{(1)} = 0.02$$

Round 2 (after rationale sharing)

$$\cdot p_1^{(2)} = 0.08 \quad \cdot p_2^{(2)} = 0.08 \quad \cdot p_3^{(2)} = 0.03$$



$\hat{p} = 0.06$ (Brier = 0.004) $\rightarrow$ NO



Private evidence enables self-correction.

